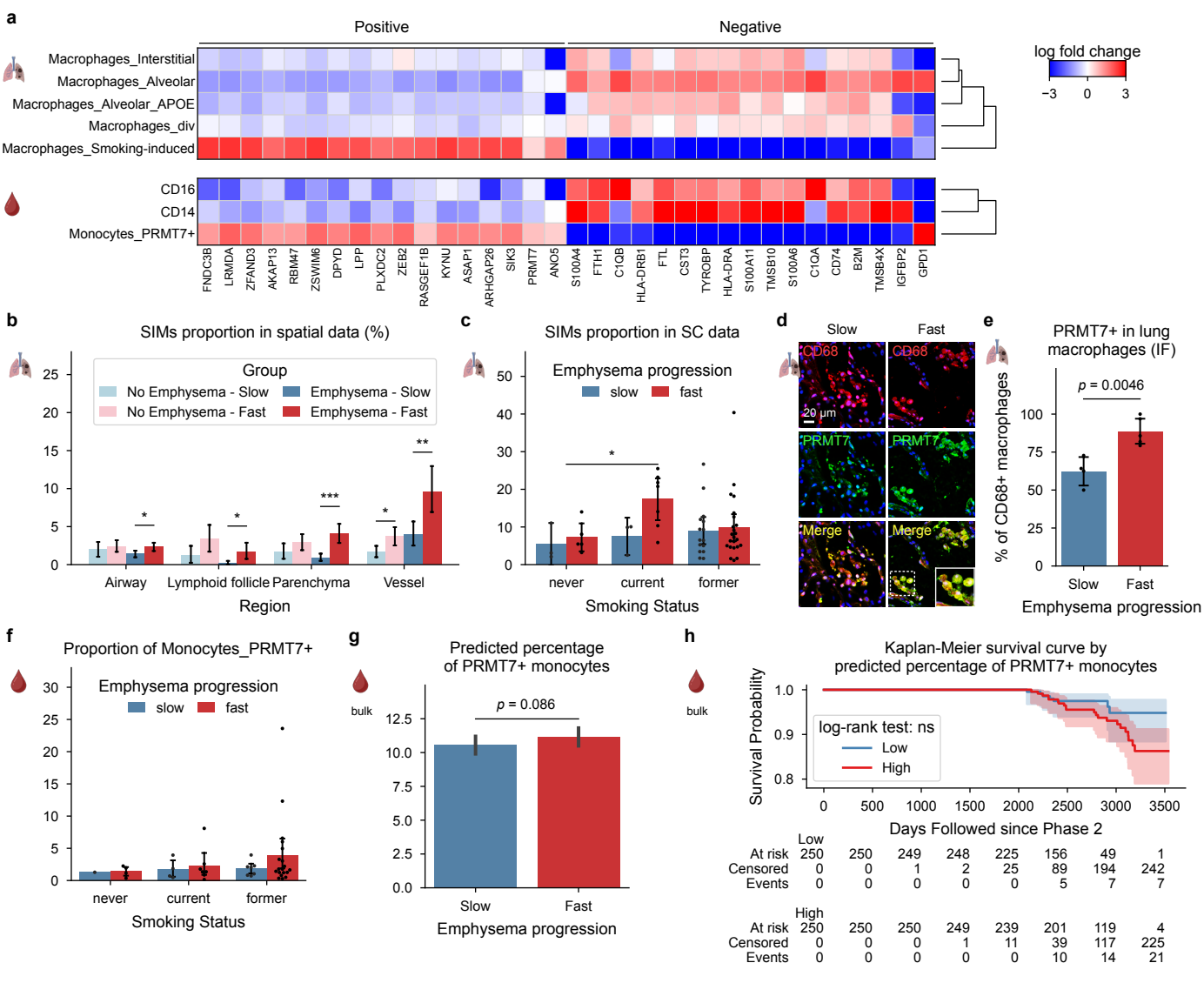

a
Positive
Negative
log fold change
−3
0
3
Macrophages_Interstitial
Macrophages_Alveolar
Macrophages_Alveolar_APOE
Macrophages_div
Macrophages_Smoking-induced
CD16
CD14
Monocytes_PRMT7+
FNDC3B
LRMDA
ZFAND3
AKAP13
RBM47
ZSWIM6
DPYD
LPP
PLXDC2
ZEB2
RASGEF1B
KYNU
ASAP1
ARHGAP26
SIK3
PRMT7
ANO5
S100A4
FTH1
C1QB
HLA-DRB1
FTL
CST3
TYROBP
HLA-DRA
S100A11
TMSB10
S100A6
C1QA
CD74
B2M
TMSB4X
IGFBP2
GPD1
b
SIMs proportion in spatial data (%)
Group
No Emphysema - Slow
Emphysema - Slow
No Emphysema - Fast
Emphysema - Fast
Airway
Lymphoid follicle
Parenchyma
Vessel
Region
c
SIMs proportion in SC data
Emphysema progression
slow
fast
never
current
former
Smoking Status
d
Slow
Fast
CD68
PRMT7
Merge
20 μm
e
PRMT7+ in lung macrophages (IF)
p = 0.0046
% of CD68+ macrophages
Slow
Fast
Emphysema progression
f
Proportion of Monocytes_PRMT7+
Emphysema progression
slow
fast
never
current
former
Smoking Status
g
bulk
Predicted percentage of PRMT7+ monocytes
p = 0.086
Slow
Fast
Emphysema progression
h
bulk
Kaplan-Meier survival curve by predicted percentage of PRMT7+ monocytes
Survival Probability
log-rank test: ns
Low
High
Days Followed since Phase 2
Low
At risk 250 250 249 248 225 156 49 1
Censored 0 0 1 2 25 89 194 242
Events 0 0 0 0 0 5 7 7
High
At risk 250 250 250 249 239 201 119 4
Censored 0 0 0 1 11 39 117 225
Events 0 0 0 0 0 10 14 21